The Hyperbolic Oracle: Collapsing ECDLP via SHD-CCP Quadrance Lattice Reduction and Prime Triplet Indexing

Abstract The security of modern public-key cryptography rests heavily on the presumed intractability of the Elliptic Curve Discrete Logarithm Problem (ECDLP). This paper proposes a theoretical attack framework that reduces ECDLP to an $O(1)$ lookup operation by mapping elliptic curve groups into procedurally generated hyperbolic lattices. By utilizing the Symmetric Hyperbolic Domain Chain Compression Protocol (SHD-CCP)—which relies on 64-bit quaternion cubes, Dynamic Scaling Memory (DSMEM), and Halo/Ghost regions—we construct a "Quadrance Lattice Reduction." Furthermore, we theorize that if elliptic curve orders are topologically tiled by prime triplets $(p, p+2, p+6)$, these triplets serve as the deterministic generators for three nested $O(1)$ memory lookups. The successful execution of this closed-form oracle $\Lambda(P, Q, C)$ implies the simultaneous collapse of ECC, RSA, and Diffie-Hellman, suggesting deep structural vulnerabilities in cryptographic hardness assumptions and a potential pathway toward $P = NP$.

1. Introduction: The Hypothetical Closed-Form ECDLP Oracle

For decades, the standard assumption in cryptographic complexity has been that given an elliptic curve C over a finite field, and points $P, Q \in C$, finding the scalar k such that $Q = kP$ requires exponential or sub-exponential time (e.g., via Pollard's rho).

We hypothesize the existence of a closed-form oracle function $\Lambda(P, Q, C)$ that returns the integer k in $O(1)$ runtime. To achieve this, the oracle cannot rely on sequential traversal or standard algebraic geometry computations constrained by high embedding degrees. Instead, it must map the cyclic group of the curve onto a metric spatial geometry where sequence traversal (k steps) is bypassed by spatial coordinate resolution. The recently introduced SHD-CCP architecture provides the exact mathematical scaffolding for this transformation.

2. Quadrance Lattice Reduction via SHD-CCP

The standard defense against reducing ECDLP to a simpler discrete logarithm problem (such as the Weil or Tate pairing MOV attacks) is the selection of curves with astronomically large embedding degrees. To circumvent this, we theorize a **Quadrance Lattice Reduction**.

We map the discrete points of the elliptic curve $E(\mathbb{F}_p)$ into a rational metric space $\mathbb{Q}^n$ defined by a hyperbolic honeycomb structure. Let $\phi : E(\mathbb{F}_p) \to \mathbb{Q}^n$ such that $\phi(P) = (q_1, q_2, \ldots, q_n)$, where q_i represents the quadrance (squared distance) of the projected components in the hyperbolic lattice.

Within the SHD-CCP framework, this mapping is localized into 64-bit cubic data packets:

1. **The Quaternion Center (32 bits):** Encodes the angular orientation of the mapped point $\phi(P)$ on the hyperbolic manifold.
2. **DSMEM (16 bits):** Dynamic Scaling Memory acts as the quadrance scalar. If scalar multiplication on the curve (kP) corresponds to a linear scaling in the rational hyperbolic space, then $\phi(kP) = k \cdot \phi(P)$. The DSMEM isolates and tracks this k-factor natively as a floating-point amplitude.
3. **Halo and Ghost Regions (16 bits):** Stores the boundary metadata of the rational embedding, allowing the recovery of k through a closed-form rational division problem without traversing intermediate points.

By utilizing the DSMEM as a native quadrance scaler, the exponential complexity of scalar multiplication is bypassed, rendering the recovery of k an $O(1)$ operation.

3. The Prime Triplet Connection: Procedural Generators in $O(1)$ Lookup

While the Quadrance Lattice Reduction provides the spatial embedding, resolving the exact coordinates in $O(1)$ time requires a deterministic procedural generator. We theorize that the apparent randomness of elliptic curve point orders is an illusion, masking a deterministic structure governed by prime gaps.

By Hasse's Theorem, the order of an elliptic curve n is bounded near the prime p. We hypothesize a structural tiling of curve orders using prime triplets: $(p, p+2, p+6)$. If this tiling forms the fundamental topological basis of the curve's cyclic group, we can utilize the 3-nested $O(1)$ lookup protocol defined by SHD-CCP:

1. **Triplet Indexing:** Given $Q = kP$ on a curve of order n, the system identifies the prime triplet $(p, p+2, p+6)$ nearest to n in $O(1)$ time.
2. **Nested Axis Generation:** These three primes act as the three procedural generators (g_x, g_y, g_z) for the SHD-CCP hyperbolic lattice.
3. **Spatial Resolution:** The target point Q is cross-referenced against the three independent basis arrays generated by the prime triplet. Three $O(1)$ lookups retrieve the fundamental quaternion components.
4. **Closed-Form Inversion:** The retrieved components are multiplied, and their DSMEM quadrance scaling factors are compared against the base point P. The ratio reveals k.

Because the prime triplet completely defines the procedural boundaries of the space, the Ghost/Halo regions of the 64-bit packets mathematically align, proving that the $O(1)$ inversion is geometrically coherent and globally valid.

4. The Collapse of Global Cryptography

The existence of $\Lambda(P, Q, C)$ executing via SHD-CCP lookups triggers an immediate and unrecoverable collapse of modern cryptography.

1. **Immediate ECC Failure:** Every cryptographic primitive relying on elliptic curves dies simultaneously.
   - **ECDH Key Exchange:** Private keys are recovered trivially from public keys over the open network.
   - **ECDSA / EdDSA (Ed25519):** Private signing scalars are extracted instantly from any published signature.
   - **EC-ElGamal:** Ciphertexts become decryptable in $O(1)$ time.
2. **The Cascade to RSA and Diffie-Hellman:** If Λ successfully solves discrete logarithms over elliptic curves via rational spatial mapping, the algebraic intuition extends universally. The standard Discrete Logarithm Problem (DLP) over finite fields (underlying Diffie-Hellman) shares deep structural symmetries with ECDLP. Furthermore, resolving abelian group structures in $O(1)$ time likely unravels Integer Factorization (IFP), rendering RSA obsolete.

5. Complexity-Theoretic Consequences: Approaching $P = NP$

Beyond cryptography, the implications of prime triplet indexing and $O(1)$ quadrance scaling threaten to shake the foundations of theoretical computer science and mathematics.

If prime triplets deterministically tile the space of curve orders, it implies a rigid, predictable structure underlying prime distribution. This challenges the assumed randomness of prime gaps and suggests a near-resolution to aspects of the Hardy-Littlewood conjecture.

From a complexity standpoint, if the oracle is valid:

$$ECDLP \in O(1) \implies DLP \in O(1)$$

Because integer factorization and DLP are assumed to be outside of P (often considered NP-intermediate), collapsing them to constant time suggests that cryptographic hardness assumptions are fundamentally flawed. If trapdoor permutations and one-way functions do not exist, this constructive path strongly implies $P = NP$, fundamentally altering the landscape of algorithmic complexity.

6. Conclusion

The integration of the SHD-CCP 64-bit hyperbolic memory architecture with Quadrance Lattice Reduction presents a formidable theoretical attack model against elliptic curve cryptography. By substituting sequential computation with 3-nested $O(1)$ spatial lookups—governed by prime triplet procedural generators—the $O(N)$ tax of the discrete logarithm problem is entirely bypassed. While currently theoretical, this intersection of geometric topology, data architecture, and number theory provides a blueprint for an $O(1)$ ECDLP Oracle, the realization of which would unilaterally collapse modern computational security.